

FMCW MIMO Radar System for Frequency-Division Multiple TX-Beamforming

Clemens Pfeffer, *Student Member, IEEE*, Reinhard Feger, *Member, IEEE*, Christoph Wagner, *Member, IEEE*, and Andreas Stelzer, *Member, IEEE*

Abstract—In this paper, a prototype automotive radar sensor is presented that is capable of generating simultaneously multiple transmit (TX) beams. The system is based on a four-channel 77-GHz frequency-modulated continuous-wave (FMCW) radar system. The number of beams, their radiated power, steering angle, and beam pattern can be changed adaptively. This is achieved by the utilization of orthogonal waveforms applied to different beams in combination with digital beamforming on the receive side. Key components are vector modulators in the TX path controlled by digital-to-analog converters. The performance of the system is shown in measurements focused on beam pattern, signal-to-noise ratio, and susceptibility in case of interfering targets at cross-range. Measurement results are discussed and compared to theory and simulations. Furthermore, crest factor minimization of the vector modulator's control signals is introduced and used to increase the achievable TX power, which will be also shown in measurements.

Index Terms—Position measurement, radar, radar signal processing.

I. INTRODUCTION

RADAR sensors for automotive applications at 77 GHz are available on the market for several years. They are commonly based on the frequency-modulated continuous-wave (FMCW) radar principle [1]. Their utilization in driver assistance systems increase the driver's safety and comfort [2]. Depending on the field of application, such as adaptive cruise control or parking aid, various sensors are used to meet different requirements regarding maximum distance, field of view (FOV), range and angular resolution, and signal-to-noise ratio (SNR) to name a few. As pointed out in [2], also the sensor size is one key feature for a flexible integration of the sensor into the vehicle and is mainly determined by the number of antennas and the used wavelength. Due to the high frequency, the development of

77-GHz monolithic microwave integrated circuits (MMIC) has led to a miniaturization of the sensors and offers a large bandwidth up to several GHz, which results in a range resolution down to centimeters.

Modern radar sensors employ beamforming techniques, not only to obtain angular information about the targets but also to narrow the transmit (TX) beam. In case of TX beamforming, only the interesting FOV is illuminated, which improves the energy efficiency. There are various techniques to achieve this, such as a dielectrical lens (LRR3 sensor by Bosch), mechanical scanning (ARS 300 sensor by Conti), or multi-column transmit patch array antenna (MRR sensor by Bosch) as listed in [2] and referenced therein.

In this paper, we present a realization of a prototype type frequency-division multiple TX-beamforming (FDMB) multiple-input multiple-output (MIMO) radar system in the 77-GHz band. It is based on the FMCW principle and provides multiple TX phased-array (PhA) beams. These are simultaneously transmitted, independently electronically steerable, and power-adjustable, just to name three among other features. On the receive (RX) side, digital beamforming (DBF) is used to obtain additional angular information within the illuminated areas. Compared to state-of-the-art sensors, the number of beams and their radiated power can be changed adaptively to optimize the sensors performance in varying scenarios. Regardless of the number of beams, only one radar measurement is required to collect the complete target scenario information. This is achieved by transmitting all beams simultaneously using the same PhA antenna. This approach and the use of RX DBF reduces the number of radio frequency (RF) components compared to subarray (SA) systems ([3], [4], and [5]). Thus, the presented approach is easier to apply in automotive applications due to reduced number of TX antennas.

In [6] we presented a realization of a 77-GHz frequency-division multiple access (FDMA) based MIMO FMCW radar system. There, different frequency offsets associated to multiple TX channels were applied to the transmitted signals, a special case of orthogonal waveforms [7]. The assignment of TX and RX channel is performed according to the offset frequency in the baseband. Compared to [6], where offset frequencies are associated with certain TX channels, offset frequencies in this work are associated to certain beams [8]. In this paper the presented system is thus an extension of [9]. There, an FDMA MIMO radar (wide FOV, short range) is simultaneously combined with one single TX PhA beam (narrow beam, long range). The novelty of the current work is that the system is able to simultaneously track multiple targets directing the beams

Manuscript received May 23, 2013; revised October 18, 2013; accepted October 21, 2013. Date of publication November 19, 2013; date of current version December 02, 2013. This work was supported in part by the Federal Ministry of Economy, Family and Youth, in part by the National Foundation for Research, Technology and Development, in part by the Austrian Research Promotion Agency, and in part by the Christian Doppler Research Association, DICE GmbH & Co KG as part of the Infineon group for funding the work carried out at the Christian Doppler Laboratory for Integrated Radar Sensors.

C. Pfeffer, R. Feger, and A. Stelzer are with the Christian Doppler Laboratory for Integrated Radar Sensors, Institute for Communications Engineering and RF-Systems, Johannes Kepler University Linz, 4040 Linz, Austria (e-mail: c.pfeffer@nthfs.jku.at; r.feger@nthfs.jku.at; a.stelzer@nthfs.jku.at).

C. Wagner is with DICE GmbH & Co KG, 4040 Linz, Austria (e-mail: christoph.wagner@infineon.com).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TMTT.2013.2287675

into the targets' directions. In the presented system, tracking is performed by illuminating predefined regions with a narrow beam and obtaining additional angular information within this beam using RX DBF. This approach accounts for typical target scenarios in the automotive sector and keeps the system complexity low. A key component is a vector modulator (VM) in each TX path of the PhA, which is used as single side-band (SSB) mixer and is controlled by digital-to-analog converters (DACs). In comparison to approaches that employ multiple voltage controlled oscillators (VCOs) and phase locked loops (PLLs) [10] to realize frequency offsets, the combination of VM and DAC offers additional degrees of freedom in modulation. This approach supports the use of different complex excitations—i.e., defining the beam pattern—at different offset frequencies at the same time. The presented concept is based on the assumption that the increase of bandwidth caused by the modulation is negligible compared to the available bandwidth. Thus, compared to frequency diverse MIMO radar systems as shown in [11] and referenced therein, in this work frequency offsets are not used to create a range-dependent beam pattern, they are used to separate different beams. The fact that every TX beam and its corresponding DBF region requires additional bandwidth in the baseband can be seen as one of the most significant drawbacks of the presented approach. Against this it may be argued that the rapid development of baseband components—analogue-to-digital converters (ADCs) and DACs are meanwhile available at sample rates in the GHz-range—and their future integration in MMICs offer enough bandwidth, especially in automotive radar applications.

This paper is divided into four parts. We briefly describe the essential system basics in Section II, which are required for the theory presented in Section III. There, the underlying signal model is presented and maximization of the transmitted power using crest factor (CF) minimization will be discussed. In Section IV, we describe the hardware that is used for the measurements presented in Section V. Finally, in Section V, we explain a calibration method and compare measurement results to simulations.

II. SYSTEM BASICS

Radar sensors illuminate the interesting FOV for locating multiple targets. The blue region in Fig. 1 marks the FOV achieved by a broad beam as for example used in classical MIMO radar systems [12]. As it can be seen, the received power varies greatly from one target to another according to the radar cross section (RCS) and distance to the sensor (illustrated by the blue waves' amplitudes). Targets with a huge RCS can cause saturation at the receiver, such as, for example a truck located in proximity to the sensor. However, the radiated power cannot be reduced because high power is required to detect far away obstacles or objects with a small RCS like persons. Introducing multiple narrow beams can resolve this problem. Each beam is steered to a certain target. The TX power of the beams can be adjusted in a way that the received power of different targets is equalized. This is advantageous as it can lower the requirements concerning the dynamic range of the receivers or the resolution of ADCs. In automotive applications, a typical target scenario as depicted in Fig. 1 requires only a

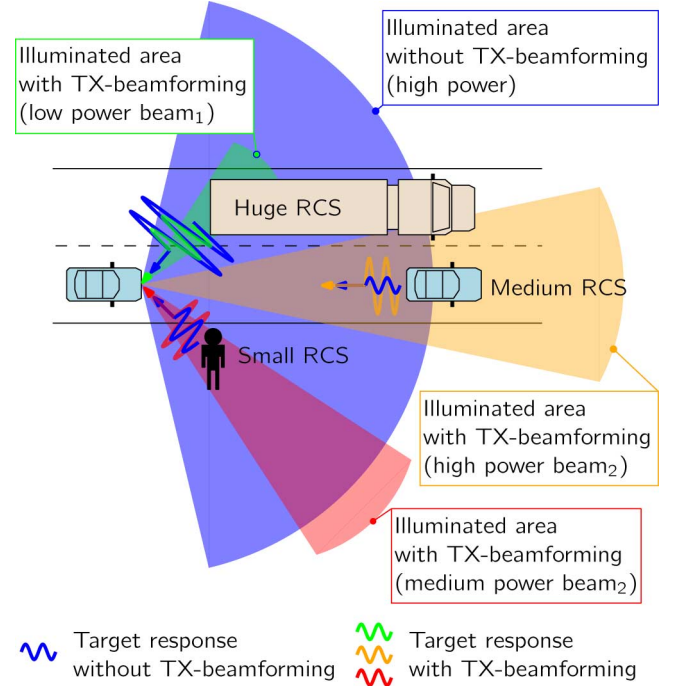


Fig. 1. Blue region marks FOV achieved with an antenna for broad azimuth coverage. In comparison to multiple independent power adjustable narrow beams (green, orange, and red) the received power varies greatly from one target to another according to the RCS and the distance to the sensor (illustrated by the blue waves' amplitude).

limited number of beams. One for detecting cars driving on the same lane, which should be steerable in case of driving through curves, and additional beams to cover the surrounding area.

There are many possible ways to realize a beamforming radar system with K independent beams. In Fig. 2(a) K analog sub-systems for analog beamforming on the TX as well as on the RX side require a large number of components. A key issue for automotive radar systems as already mentioned in Section I is the given restricted size of the sensor. Therefore, a solution as depicted in Fig. 2(b) reduces the number of components and the complexity of the whole system. Analog beamforming on the RX side is replaced by DBF and the K different TX beams are transmitted sequentially. The time schedule for transmitting the K different beams leads to an increased overall measurement time proportional to K . This also complicates the practical application of such a system in moving target scenarios. To overcome this issue in this paper, the different beams are separated using K different TX frequencies $f_P[k]$ ($k=1, 2, \dots, K$) associated to the beam pattern as shown in Fig. 2(c). Range and angular position can be obtained using standard methods of DBF after performing K digital down-conversions of the received signals according to $f_P[k]$. Of course, the reduction of the number of required antennas by utilization of TRX channels is not limited to the presented system shown in Fig. 2(c).

In this work we use a uniformly excited linear array, to exemplify the basic principle, considering a $\lambda/2$ uniformly spaced linear array (ULA) with $M_t = 4$ TX antennas and colocated $M_r = 4$ RX antennas [13] realized using TRX cells [14], where λ is the wavelength of the transmitted signal. Therefore, a radar system as shown in Fig. 3 is used (cf.

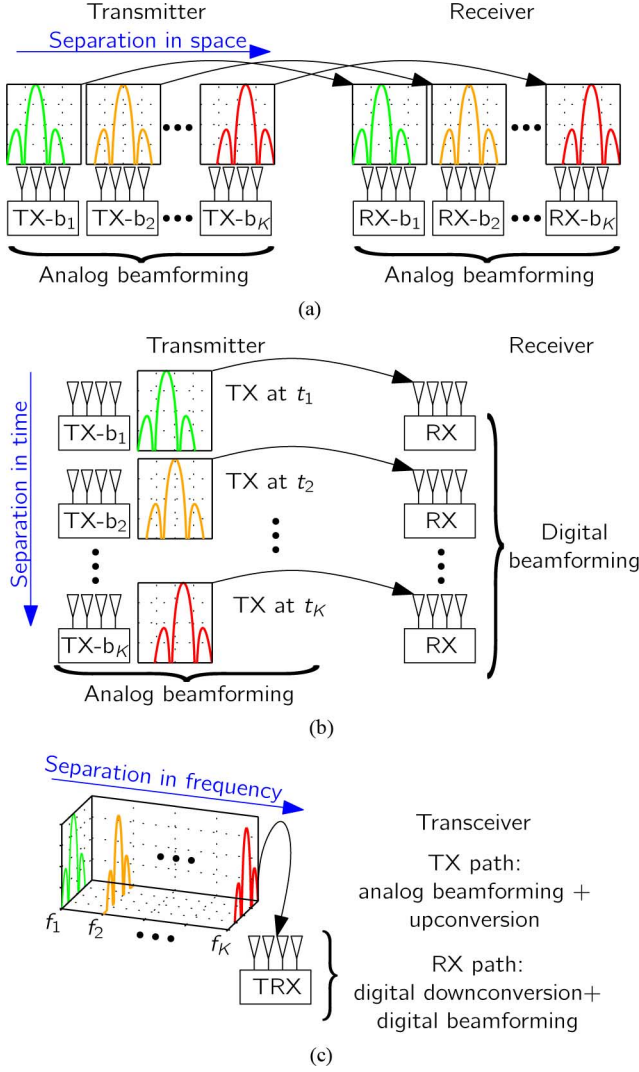


Fig. 2. Three possible realizations of a radar system with K different TX beams. (a) K analog TX and RX beamforming subsystems. (b) DBF on the RX side. K different beams are transmitted consecutively. (c) K different beams are transmitted simultaneously at K different frequencies. After digital down-conversion DBF methods can be applied.

[15]). As it is required for the FMCW principle [1], the RF signal generator linearly increases the RF signal frequency f_{RF} , according to $f_{RF} = f_0 + k_r t$, where f_0 denotes the start frequency and $k_r = B_{SW}/T_{SW}$ the ramp slope, with B_{SW} the sweep bandwidth and T_{SW} the sweep duration. This signal is distributed to the RX mixers and to the VMs. In this approach, the VMs work as SSB mixers for generating the frequency shifted output signals at frequencies $f_P[k] = f_{RF} + f_M[k]$ with respect to f_{RF} and with offset frequencies $f_M[k]$. As an example, one TX beam with steering angle 0° and uniformly excited TX beam pattern at offset frequency f_M is assumed. In this case every TX cell transmits the same frequency ramp $f_P = f_{RF} + f_M$. Steering this beam is achieved by changing the phase relations between the TX cells, which is performed in baseband applying phase offsets to the complex control signals of each VM. In case of multiple beams, the complex control signals are calculated by simply summing up of the required control signals of each TX beam for each VM. Furthermore, the complex excitations are applied defining the different desired

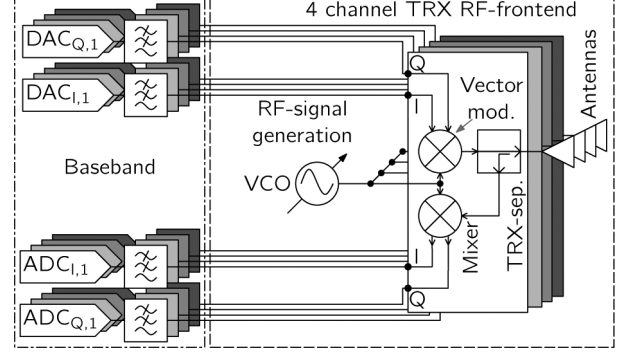


Fig. 3. Block diagram of the used FDMB radar system with four TRX channels and VMs in the TX path.

TX beam patterns. The RX mixers convert the received signals to zero-IF. The mixer's output signals are filtered and digitized for digital down-conversion and signal processing.

The following example further illustrates the proposed principle of TX beamforming for generating multiple beams. Assume that a target is located at a certain range r and at a certain angle θ_T . The steering angles of three beams are set to -30° , 0° , and 30° , equally radiated power of the multiple TX beams is assumed, and offset frequencies $f_M[k]$ ($k=1, 2, 3$) 0.1, 0.2, and 0.3 are used, which are normalized to the sample rate f_s . Fig. 4(a) shows a simulated spectrum of the intermediate frequency (IF) signal of one RX channel for $\theta_T = 0^\circ$. The range information is contained in the frequency difference between $f_M[k]$ and the response of the target, which depends on the round-trip delay-time (RTDT) and the associated $f_M[k]$ [15]. As Fig. 4(a) depicts, the unambiguous range is defined by the spacing between offset frequencies or highest offset frequency and half the sample rate of the ADCs. The range as well as the angular information of targets illuminated by the corresponding TX beams are contained within the associated unambiguous ranges. The differences in received power, marked by black crosses, originate from the different TX beam patterns as shown in Fig. 4(b), where the simulated target's angular position was varied from -90° to 90° . Assuming $B_{SW}=1$ GHz, $T_{SW}=1$ ms, and a DBF region of $r_{max}=300$ m, the required spectral bandwidth in baseband B_{IF} can be calculated to $B_{IF} = (2r_{max}B_{SW})/(cT_{SW}) = 2$ MHz with c denoting the speed of light. Depending on the target scenario and the required TX power it is possible to reduce or increase B_{IF} for each beam adaptively.

The presented approach can be interpreted as a variation of the so called subarray-MIMO (SA-MIMO) as discussed in [4]. Instead of physically realized subarrays transmitting multiple beams simultaneously, frequency offsets are used to separate them. This reduces the overall size of the radar sensor and enables coherent signal processing. As pointed out in [4], MIMO radar systems illuminate a large volume of space, which can be used for search, but not for track mode. In the presented paper, the possibility to build an angular and range dependent fence of beams covering only the interesting FOV increases the energy efficiency. This overall TX beam pattern can be changed adaptively and accounts for typical scenarios in automotive radar applications, such as driving in the city or driving on a motorway.

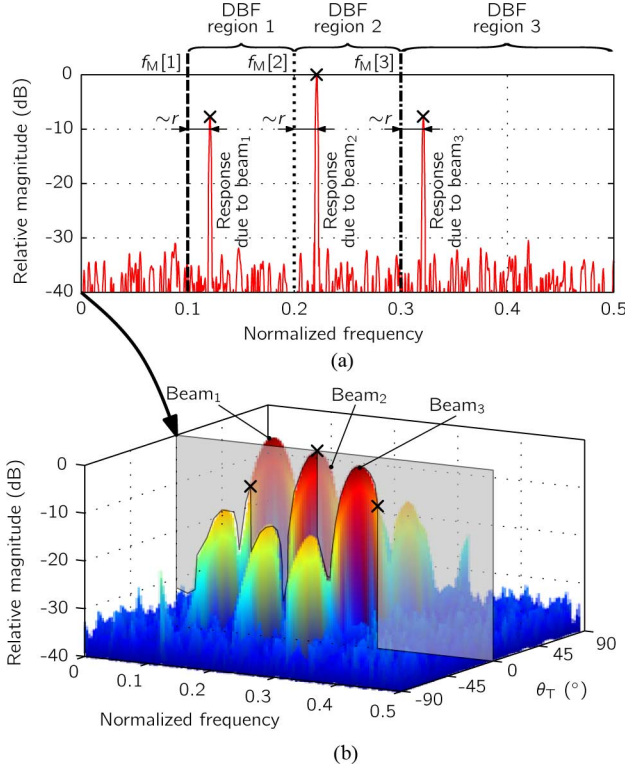


Fig. 4. Simulated IF signal spectra, of one RX channel of the FDMB radar system showing the target response caused by a single target at angle θ_T ($K = 3$, steering angles of -30° , 0° , and 30° , offset frequencies normalized to the sample rate of 0.1, 0.2, and 0.3, beams with equal radiated power). (a) Spectrum for $\theta_T = 0^\circ$ with marked unambiguous ranges associated with the corresponding beams. (b) Spectra for varying $\theta_T = -90^\circ$ to 90° .

Compared to airborne radar systems in automotive radar applications it is hard to distinguish between search and track mode as the target scenario can change in a fraction of a second. As a consequence, an automotive radar sensor has to operate mainly in search mode, in order to detect, e.g., playing children between parking cars or crossing cars. Nevertheless, the presented approach is one step towards SA-MIMO and higher energy efficiency. By using DBF on the RX side the complexity is shifted into the baseband and signal processing. The development of powerful signal processors further promote this evolution.

III. THEORY

As already mentioned in Section I, the presented system employs orthogonal waveforms, in the presented specific case different frequencies, to separate different beams. In [8], Friedlander presented a general signal model for multiple beam TX beamforming, which also includes the utilization of orthogonal waveforms. This will be adapted for the FMCW radar principle and only the necessary formulas will be stated in Section III-A. In Section III-B the technique of CF minimization is applied to the VM's control signals. The CF denotes the ratio between peak amplitude and the RMS value of a waveform. As the transmitted power is limited by the controls signals' peak amplitude level of the VMs, reducing the CF and thus the peak amplitude offers the possibility to increase the transmitted average power.

A. Signal Model

For the presented signal model the following assumptions were made (cf. [16] and [17]). The target is static and located in the array's far field, the receivers deliver complex valued, low-pass filtered, and noiseless data, and ideal RF components are assumed. In case of a single target and after TX beamforming, the signal model for the IF signals at the output of the mixer with index m_r can be written as

$$s_{\text{IF,cont}}[t, m_r] = \sum_{k=1}^K \sum_{m_t=1}^{M_t} \underbrace{A e^{-j2\pi(k_r 2r/ct + f_M[k]t)}}_{\text{frequency term}} \underbrace{e^{-j2\pi((f_0 + f_M[k])2r/c + \Delta u_t[k](m_t - 1)) + j\Phi_t}}_{\text{phase term + TX array variable}}, \quad (1)$$

with the target distance r with respect to the array reference point and a phase term Φ_t , which accounts for a system related phase and the reflection properties of the target. Amplitude influences such as the antenna gain and the RX signal amplitude caused by the RCS of the target are subsumed by A . The range information, which is assumed as sinusoidal signal with frequency $k_r 2r/c + f_M[k]$ of the corresponding beam is contained in the frequency term. The phase term contains the range dependent phase term $f_0 2r/c$, used for DBF and the TX array variable. The latter defines how much power is received due to the shape of the TX beam pattern. The TX array variable (cf. [18]) can be written as

$$\Delta u_t[k] = \underbrace{\frac{d}{\lambda[k]} \sin(\theta_T)}_{u_T} - \underbrace{\frac{d}{\lambda[k]} \sin(\theta_t[k])}_{u_t[k]} \quad (2)$$

with θ_T denoting the angle of the target relative to broadside, $\theta_t[k]$ denoting the steering angle of the k th beam and d denotes the spacing of the antennas.

1) *Approximations*: In case of modulation, the wavelength $\lambda[k] = c/(f_0 + f_M[k])$ also depends on $f_M[k]$. The assumptions $d \approx \lambda_0/2 = c/f_0/2$ and $\lambda[k] \approx \lambda_0$ simplify the further derivations. The resulting phase error, if $f_M[k]$ is neglected in (1) with $\max |\sin(\theta[k]) - \sin(\theta_t[k])| = 2$, can be stated as

$$\Delta \varphi[k, m_t] = 2\pi \left(\underbrace{\left(\frac{f_M[k]}{c} 2r \right)}_a + \underbrace{\left(\frac{f_M[k]}{f_0} (m_t - 1) \right)}_b \right). \quad (3)$$

Term a is range dependent and does not effect the shape of the TX beam pattern, as it adds the same phase shift to all transmit channels of the associated beam. Term b affects the shape of the beam pattern since it depends on m_t and has to be considered in the design of the system, such that

$$\Delta \varphi_{\max} = 2\pi \frac{f_M[k]}{f_0} (M_t - 1) \ll \Delta \varphi_{\text{des}} \quad (4)$$

where $\Delta \varphi_{\max}$ denotes the maximum phase error between the first and last array element. In (4) $\Delta \varphi_{\text{des}}$ denotes the minimum allowable phase error to achieve the desired system performance. In the presented system the limit is the achievable angular accuracy of the VMs, which is in the range of a few degrees. For $f_M = 5$ MHz, $f_0 = 77$ GHz, and $M_t = 4$ the maximum phase error

is $\Delta\varphi_{\max}=0.07^\circ$ and can thus be neglected. As a result, (1) can be approximated by

$$s_{\text{IF,cont}}[t, m_r] = \sum_{k=1}^K \sum_{m_t=1}^{M_t} A e^{-j2\pi(f_0 2r/c + u_t[k](m_t-1) + k_r 2r/ct + f_M[k]t) + j\Phi_t}. \quad (5)$$

If (4) is fulfilled for a given radar system, the TX beam pattern can be designed without taking the modulation frequencies into account.

2) *Digital Beamforming*: In the next step a discrete-time Fourier transform (DTFT) is applied to (5). To account for discrete sampling, the continuous time t in (5) is substituted by $t=(n-1)/f_s$ with n denoting the discrete-time index where $n=1, 2, \dots, N$ and N denoting the number of samples. Calculating the DTFT leads to

$$S_{\text{IF}}[f, m_r] = \sum_{n=1}^N s_{\text{IF,cont}}\left[\frac{n-1}{f_s}, m_r\right] e^{-j2\pi f/f_s(n-1)}. \quad (6)$$

As already mentioned in Section II, the beams are separated for further calculations by digital down-conversion of (6) according to $f_M[k]$. In order to limit the down-converted signals to the unambiguous range, a low-pass filter $H_{\text{LP}}[f, k]$ with cutoff frequency $\leq \Delta f_M[k]$ is used. The value $\Delta f_M[k]$ is calculated as the difference between two neighboring modulation frequencies. For uniformly spaced $f_M[k]$, the cutoff frequency is constant and the down-converted IF signals can be written as

$$\check{S}_{\text{IF,DDC}}[f, m_r, k] = H_{\text{LP}}[f, k] S_{\text{IF}}[f - f_M[k], m_r]. \quad (7)$$

Using the relation $r = f_s c / (2k_r) f$, (7) can be scaled to range r and is denoted with $S_{\text{IF,DDC}}[r, m_r, k]$. Then, a delay-and-sum beamformer is applied in the next step as presented in [16]. The DBF output signals for the k th beam can be written as

$$S_{\text{IF,DBF}}[r, u_r, k] = \sum_{m_r=1}^{M_r} S_{\text{IF,DDC}}[r, m_r, k] e^{-j2\pi u_r(m_r-1)} \quad (8)$$

with

$$u_r = \frac{d}{\lambda} (-\sin(\theta_{\text{DBF}})) \quad (9)$$

and θ_{DBF} denoting the look direction. For further considerations, complex valued spatial weighting factors $w_t[m_t, k]$ for analog TX are introduced, which can be written as

$$w_t[m_t, k] = v[k] a_t[m_t, k] e^{-j2\pi\varphi_t[k, m_t]}. \quad (10)$$

In (10) $a_t[m_t, k] \in \mathbb{R}$ denotes the excitation coefficients (with $0 < a_t[m_t, k] \leq 1$), $\varphi_t[k, m_t] = -u_t[k](m_t-1)$ the required phase settings, and $v[k] \in \mathbb{R}$ the beam weighting factor, which distributes the TX power between the K beams. In this paper, the focus is on uniformly excited TX beam pattern and therefore

$a_t[m_t, k] = 1$. The required complex valued control signals of the VMs are generated by DACs and can be written as

$$s_M[t, m_t] = A_{\text{VM}}^{\text{LR}} \sum_{k=1}^K \left\{ w_t[m_t, k] e^{j(2\pi f_M[k]t)} \right\}. \quad (11)$$

Here, the maximum full scale amplitude $A_{\text{VM}}^{\text{LR}}$ of the VM's control signals for linear behavior restricts the free choice of $w_t[m_t, k]$ such that

$$\sum_{k=1}^K v[k] \leq 1 \quad (12)$$

must be fulfilled. It should be noted, that this condition is based on the assumption that T_{SW} is larger than at least one period of the envelope of $s_M[t, m_t]$.

B. Crest Factor Minimization

The control signals of the VMs (11), represent periodic multitone signals with K complex exponentials. Techniques to adjust the phases $\varphi_{\text{CF}}[k]$ of the sinusoidal tones with $f_M[k]$, such that for a given power spectrum the peak value of the amplitude $s_M[t, m_t]$ is minimized are referred to as CF minimization and have been subject of research for a long time [19]–[21]. In this work the aim is to reduce the peak levels of $s_M[t, m_t]$ for given values of K , $v[k]$, and $w_t[m_t, k]$ in order to provide an additional gain a_{CF} , which can be applied to the TX signals without exceeding the linear region of the VMs.

First, (11) is rewritten as

$$s_M[n, m_t, \varphi_{\text{CF}}[k]] = A_{\text{VM}}^{\text{LR}} \sum_{k=1}^K \left\{ w_t[m_t, k] e^{j(2\pi f_M[k]/f_s(n-1) + \varphi_{\text{CF}}[k])} \right\}, \quad (13)$$

to account for time-discrete sampling of the DACs with a sample rate of f_s . $\varphi_{\text{CF}}[k]$ represents freely choosable phase values associated to $f_M[k]$ of a certain beam pattern. There exists no analytic closed-form method and thus a search algorithm is used as option to obtain the phase values for a given number of beams with constant $v[k]$ and constant steering vectors. The search algorithm substitutes random values $\tilde{\varphi}_{\text{CF}}[k]$ for $\varphi_{\text{CF}}[k]$ into (13) and calculates the CF of the complex signal $s_M[n, m_t, \tilde{\varphi}_{\text{CF}}[k]]$. The quotient in complex amplitude between $s_M[n, m_t, \varphi_{\text{CF}}[k]]$ with $\varphi_{\text{CF}}[k] = 0 \forall k$ and CF optimized $s_M[n, m_t, \tilde{\varphi}_{\text{CF,opt}}[k]]$, where $\tilde{\varphi}_{\text{CF,opt}}$ denotes the phase values for the lowest CF, can be applied as additional gain a_{CF} and $w_t[m_t, k]$ in (13) is written as

$$w[m_t, k] = a_{\text{CF}} v[k] w_t[m_t, k]. \quad (14)$$

In automotive applications with a limited number of beams and a manageable number of different settings concerning the steering angles and the TX power distribution between the beams, look-up tables with precalculated values of $\varphi_{\text{CF}}[k]$ can be used. This reduces the computational effort and increases the performance of the radar sensor in predefined scenarios.

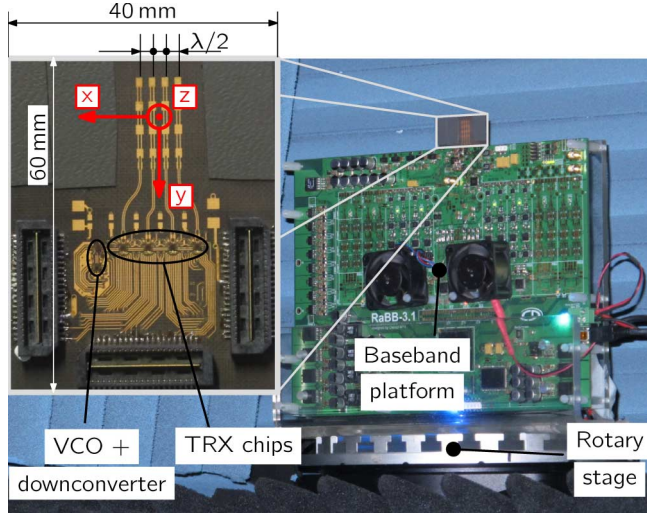


Fig. 5. Photograph of the RF frontend and the baseband board mounted on a rotary stage.

TABLE I
PARAMETERS OF THE PROTOTYPE HARDWARE.

TX power	10 dBm EIRP
antenna gain	12.8 dBi
bandwidth B_{SW}	2 GHz
start frequency f_0	76 GHz
ramp duration T_{SW}	1 ms
sampling rate ADC, DAC	20 MS/s
ADC, DAC resolution	14 bit
IF gain	15 dB
ADC full scale	2 V _{PP}
DAC full scale	20 mA _{PP}

IV. PROTOTYPE HARDWARE

The prototype radar system is based on a 77-GHz RF frontend with four TRX channels in combination with a baseband board. As can be seen in Fig. 5, the RF frontend, which was also used in [14], [15], [22], and [23], has four $\lambda/2$ spaced differentially fed patch antennas [24]. They are connected to TRX chips with integrated VMs [14]. The RF signal generation is realized by a VCO in combination with a down-converter and an offset PLL concept as described in [25]. All other components, such as the PLLs, ADCs, DACs, analog filters, and the power supply, are located on the baseband board described in [26]. It was designed to provide a reusable radar evaluation platform that is used in combination with interchangeable RF frontends. A Field Programmable Gate Array (FPGA) controls the sampling process and manages the data transfer to the computer via a USB 2.0 interface. Table I summarizes the hardware parameters.

V. MEASUREMENTS

First, a calibration method for the presented radar system is briefly described. Then measured beam patterns are compared with simulations. All beam pattern measurements were conducted without the use of an external measurement receiver. The mean absolute output power of the receive mixer of a single RX channel is used as an estimate for the received power in conjunction with a reflecting target. This method for estimating the so called two-way beam pattern was also used in [2]. Additional measurements show the impact on the received power using

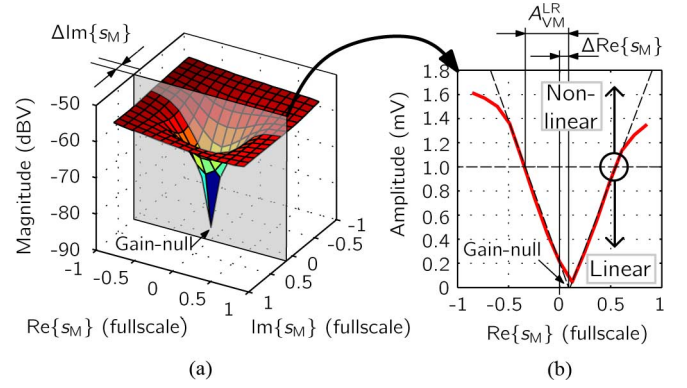


Fig. 6. Calibration of gain-null offsets $\Delta \text{Re}\{s_M[m_t]\}$ and $\Delta \text{Im}\{s_M[m_t]\}$. (a) Received power of a calibration target for control signals $\text{Re}\{s_M[m_t]\}$ and $\text{Im}\{s_M[m_t]\}$ set between -0.8 fullscale and 0.8 fullscale. (b) Received amplitude vs. $\text{Im}\{s_M[m_t]\}$ to estimate A_{VM}^{LR} for linear operation of VM.

CF minimization in order to increase the average transmitted power. Finally, the proposed radar system is compared with an FDMA based MIMO FMCW radar system, which is realized with the same hardware and does not employ TX beamforming. Differences between them, concerning SNR and susceptibility to interfering targets in case of angle estimation are discussed. All measurements have been carried out in a semi-anechoic chamber. Corner reflectors CR_A , CR_B , CR_C with RCSs of 10.5 dBsm, -1.5 dBsm, and -13.5 dBsm are used as targets to reflect the transmitted signal.

A. Calibration

In the presented FDMB MIMO system, the DAC controlled VM is the key component. Even if the VM itself is calibrated, different line lengths of the feeding networks on the printed circuit board (PCB) add unknown phase terms. Therefore, calibration of the whole system is of essential importance for the correct function. A common method is to place a well known target at broadside and estimate the unknown terms, such as phase and amplitude from radar measurements. Fig. 6(a) shows the received power of a target CR_A at broadside. The DC control signals $\text{Re}\{s_M[m_t]\}$ and $\text{Im}\{s_M[m_t]\}$ of one VM were set to varying values from -0.8 fullscale to 0.8 fullscale. Amplitude and phase are estimated from the target response at each setting of the VM. All other TX channels are disabled by turning off the RF power amplifier. As one can see, the gain-null point, which serves as the origin of the gain vector, does not correspond to the center of the control signal region. In case of sinusoidal modulation, the offsets $\Delta \text{Re}\{s_M[m_t]\}$ and $\Delta \text{Im}\{s_M[m_t]\}$ cause an additional leakage signal [23]. Therefore, a precorrection is applied to the VM's control signals. To avoid distortions caused by nonlinearities of the VM, the maximum amplitude A_{VM}^{LR} has to be restricted to the linear operating range of the VM as shown in Fig. 6(b). This procedure is repeated for all other TX channels. The recorded phase informations are used to calibrate the TX-RX combinations. It is obvious that this simple calibration cannot be used for online calibration or recalibration during sensor operation. A possible future solution would be an integration of an on-chip built-in-self-test as for example presented in [27]. The collected calibration data was used during all following measurements.

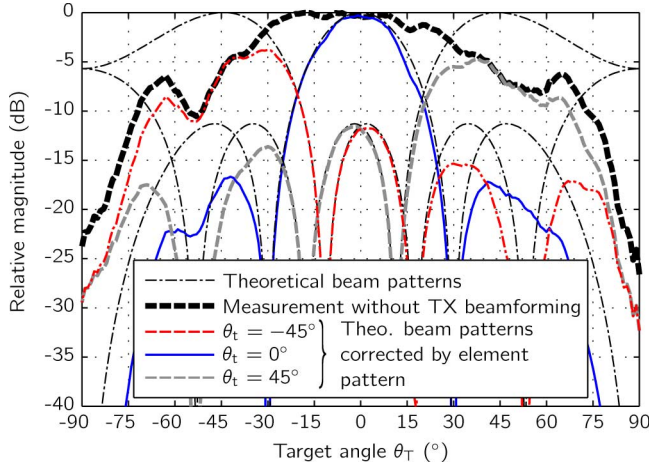


Fig. 7. Theoretical uniform beam pattern (black dashed dotted) for steering angles θ_t of -45° , 0° , and 45° ($M_t = 4$ and $\lambda/2$ spaced TX antennas). Measured two-way beam pattern of a single antenna array element (dashed black) and resulting modified theoretical patterns (dashed dotted red, blue, dashed gray) of beams steered to -45° , 0° , and 45° .

B. Beam Pattern

TX beam pattern measurements are used to verify the accuracy of the calibration and to compare the achieved performance with theory. Therefore, a target CR_A is set to broadside and remains unchanged at this position during the following measurements. The radar system is rotated from angles of -90° to 90° in steps of 1° . θ_T denotes the relative angle between z-axis perpendicular to the RF-frontend surface and the geometric connection line between the array reference point and CR_A . Coherent averaging of the sampled output signal of one RX mixer over 100 FMCW measurements at every angular step is performed in order to improve the SNR. The mean received power of the target response is then estimated using the fast Fourier transform (FFT). The antenna characteristic determined using this method is referred to as two-way antenna beam pattern ([2] and [4]), which represents the combination of RX and TX antenna beam pattern. In the first step, a two-way beam pattern of a single antenna element is recorded. The result is shown as dashed black curve in Fig. 7. It is in good agreement with results carried out in [24], noting, that therein an external measurement receiver was used. The decrease in gain near $\pm 53^\circ$ is mainly caused by radiation of surface waves (cf. [24]) and has to be considered in the simulation of the beam patterns, as it affects the TX beam pattern results. The obtained two-way single element beam pattern is used together with standard theoretical TX beam pattern theory, which is based on isotropic radiators to formulate a model that allows to verify the accuracy of the following TX beam pattern measurements. In the following, uniform excitations $a_t[m_t, k] = 1$ will be used. The black dashed dotted lines in Fig. 7 represent theoretical uniform TX beamforming patterns calculated using classical array theory ($M_t = 4$ and $\lambda/2$ spaced TX antennas) and the RF frequency of the radar frontend for steering angles θ_t of -45° , 0° , and 45° . The modified simulated TX beamforming patterns taking the element pattern into account are shown in Fig. 7.

In the next step $K=5$ simultaneous TX beams with equal transmitted power are generated at $f_M[k]$ ($k=1, 2, \dots, K$) of

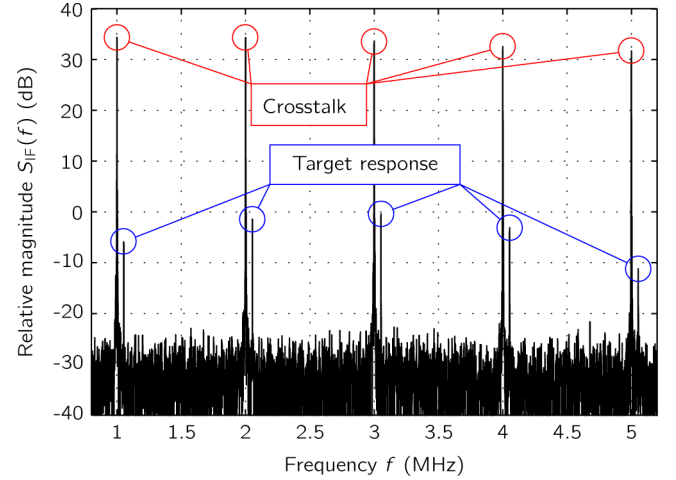


Fig. 8. IF-spectrum of one RX-channel for target at $\theta_T=0^\circ$ ($K=5$, $M_t=4$, $\lambda/2$ spaced TX antennas, offset frequencies of 1 MHz, 2 MHz, 3 MHz, 4 MHz, and 5 MHz, steering angles of 20° , 10° , 0° , -10° , and -20° , and equal radiated power).

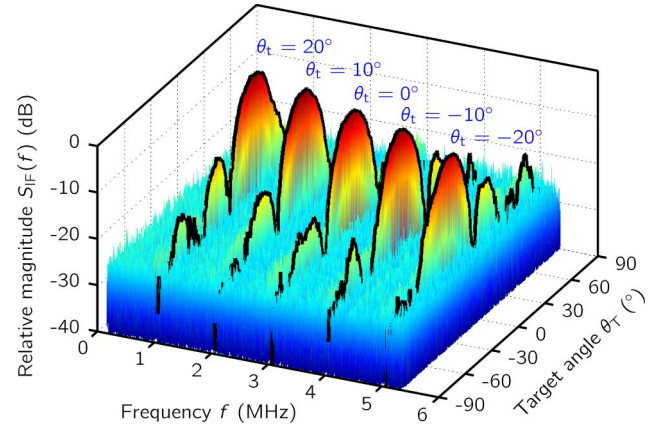


Fig. 9. Measured IF-spectra of one RX-channel for θ_T of -90° to 90° in steps of 1° . Two-way patterns are marked black obtained from target responses associated to the offset frequencies.

1 MHz, 2 MHz, 3 MHz, 4 MHz, and 5 MHz with steering angles θ_t of 20° , 10° , 0° , -10° , and -20° . The objective of this measurements is to evaluate the influence of phase errors as stated in (4) in case of neglecting the different wavelengths $\lambda[k]$ in (2) and to confirm the accuracy of TX calibration. The unambiguous range is 75 m and the resulting IF spectrum of one RX channel for $\theta_T=0^\circ$ is shown in Fig. 8. Compared to Fig. 4(a), additional tones at the offset frequencies can be observed in Fig. 8. These crosstalk signals are typical for FMCW radar systems and stem from direct coupling of TX and RX antenna, non-idealities of the rat-race coupler used for TX/RX separation, and mismatch of the antennas, just to name the most important examples in practice. To emphasize the simultaneity of the proposed principle in generating multiple beams, Fig. 9 shows the IF spectra for values of θ_T from -90° to 90° in steps of 1° collected by rotating the radar system (cf. Fig. 4(a) and (b)). The crosstalk signals are removed for clarity in this plot and the received power of the target responses are marked by black lines, which represents the two-way pattern at the associated offset frequency.

In Fig. 10 the measured two-way patterns of the five beams (θ_t

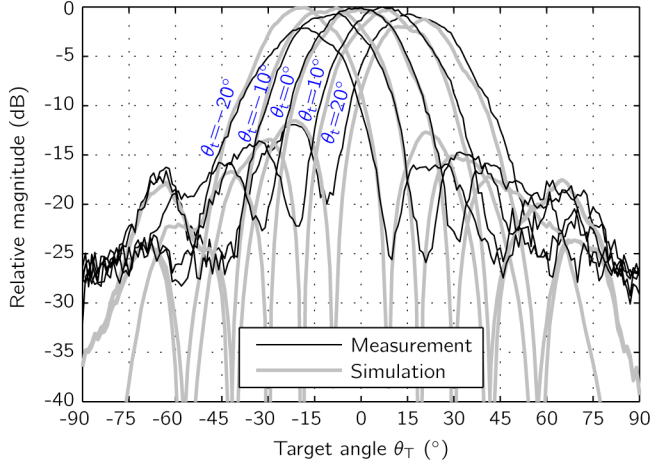


Fig. 10. Achieved uniform two-way beam patterns ($K=5$, $M_t = 4$ and $\lambda/2$ spaced TX antennas) of one measurement with steering angles of 20° , 10° , 0° , -10° , and -20° at offset frequencies 1 MHz, 2 MHz, 3 MHz, 4 MHz, and 5 MHz compared to simulation.

TABLE II
PARAMETERS USED FOR CF MINIMIZATION.

k	1	2	3	4	5	$v[k]$	a_{CF}
f_M	1 MHz	2 MHz	3 MHz	4 MHz	5 MHz		
Three beams							
θ_t	20°	0°	-20°			$1/3$	1
θ_t	20°	0°	-20°			$1/3$	1.34
φ_{CF}	199.1°	206.2°	33.3°				
Five beams							
θ_t	20°	10°	0°	-10°	-20°	$1/5$	1
θ_t	20°	10°	0°	-10°	-20°	$1/5$	1.79
φ_{CF}	258.0°	351.4°	205.1°	52.0°	21.6°		

of 20° , 10° , 0° , -10° , and -20°) are compared with the modified theoretical beam patterns taking the measured single element pattern into account. Simulation and measurement agree well (also see [14]). The reduced power level of the fifth beam ($\theta_t = -20^\circ$ and $f_M[5] = 5$ MHz) is caused by filter properties of the prototype baseband hardware. As a result, it has been shown that, neglecting different wavelengths $\lambda[k]$ in beam pattern synthesis does not lead to a detectable decrease of performance in the presented system. This measurement also shows the calibration works as expected.

C. Crest Factor

To evaluate the impact of CF minimization on the achievable increase of TX power, measurements with and without CF minimization for three, respectively five static beams were carried out. Table II lists the parameters. The algorithm described in Section III-B was used to calculate a_{CF} . In order to further illustrate the impact of CF minimization, control signals of one VM with (blue curves) and without (red curves) CF minimization in case of $K=5$ are shown in Fig. 11. As one can see from Fig. 11(a), the uncorrected control signal lies within the unity circle (A_{VM}^{LR} is normalized) according to (12) and reaches a maximum peak value of fullscale, which limits the average TX power. The same TX power is achieved with the corrected signal despite the fact that the peak amplitude value is approximately only half of the uncorrected

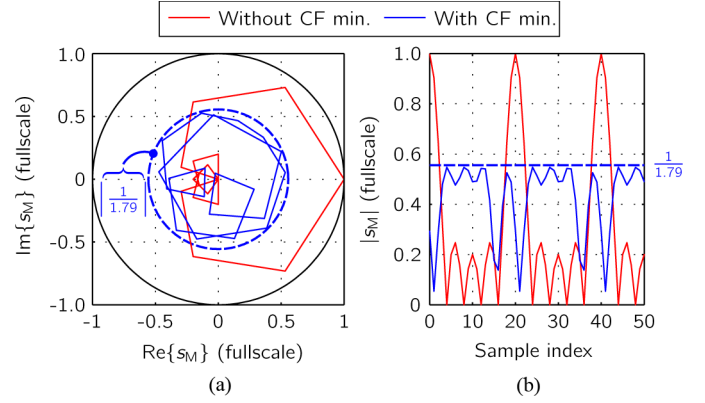


Fig. 11. Control signal s_M of one VM in case of five simultaneously generated beams with (blue) and without (red) CF minimization. (a) Constellation diagram. (b) Absolute values.

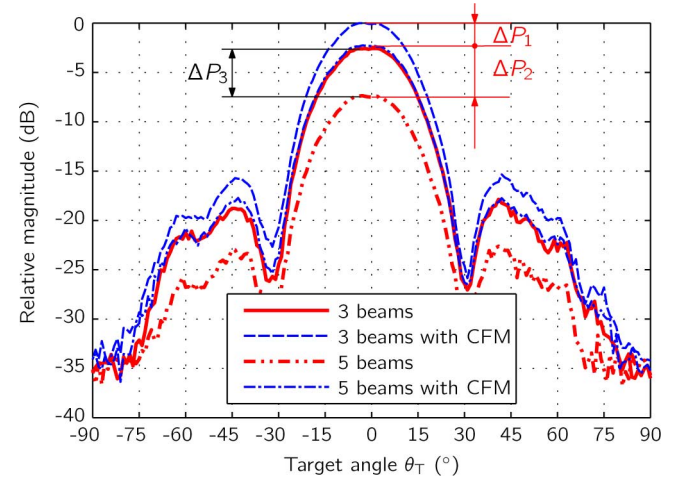


Fig. 12. Increase of TX power achieved with CF minimization in case of 3 beams of $\Delta P_1 = 2.5$ dB and 5 beams of $\Delta P_2 = 5.0$ dB ($M_t = 4$ and $\lambda/2$ spaced TX antennas). For clarity only two-way beam pattern of beams set to broadside are plotted. Magnitude difference $\Delta P_3 = 4.4$ dB caused by the different TX power distributions.

signal as shown in Fig. 11(b). Consequently, the average TX power can be increased. The increase of TX power ΔP_1 for the scenario with three beams can be calculated with $\Delta P_1 = 20 \log_{10}(a_{CF}|_{3 \text{ beams}}) = 2.5$ dB and for the scenario with five beams $\Delta P_2 = 20 \log_{10}(a_{CF}|_{5 \text{ beams}}) = 5$ dB. For clarity, only the achieved two-way beam pattern for steering angle $\theta_t = 0^\circ$ are shown in Fig. 12. The beams are normalized to the beam pattern achieved with three beams and CF minimization. The predicted values for ΔP_1 and ΔP_2 agree well with the measured values at broadside as well as the magnitude difference $\Delta P_3 = 20 \log_{10}(v[k]|_{3 \text{ beams}}/v[k]|_{5 \text{ beams}}) = 4.4$ dB caused by the different TX power distributions splitting the overall power between three respectively five beams.

D. DBF With Multiple Beams

In this section the proposed FDMB MIMO concept is compared to an FDMA based MIMO radar concept concerning SNR, angular resolution, and susceptibility of angle estimation in case of an interfering target. Common to both radar concepts

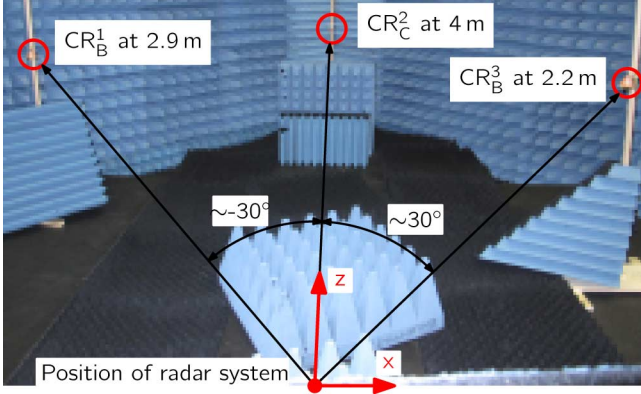


Fig. 13. Measuring arrangement for the SNR measurements. Targets of interest are CR_C^2 (-13.5 dBsm) and CR_B^3 (-1.5 dBsm).

is the utilization of frequency offsets applied to the transmitted signals and RX digital beamforming. In case of FDMB MIMO, the frequency offsets are used to separate different beams with applying TX beamforming and in FDMA MIMO to separate different TX channels without applying TX beamforming. The VM's control signals $s_M^{FDMA}[t, m_t]$ in FDMA MIMO operation can be stated as

$$s_M^{FDMA}[t, m_t] = A_{VM}^{LR} v[m_t] e^{j(2\pi f_M^{FDMA}[m_t]t)}. \quad (15)$$

In this way, no TX beamforming is applied resulting in two-way beam pattern as shown in Fig. 7 as black curve. The used frequency offsets $f_M^{FDMA}[m_t]$ ($m_t=1, 2, \dots, M_t$ and $M_t=4$) are 1 MHz, 2 MHz, 3 MHz, and 4 MHz. The maximum peak level of $s_M^{FDMA}[t, m_t]$ is equal to A_{VM}^{LR} and furthermore the condition $\sum_{m_t=1}^{M_t} v[m_t]=1$ is assumed for $s_M^{FDMA}[t, m_t]$ (cf. Equation (12)).

In the first scenario, three corner reflectors have been placed as shown in Fig. 13. CR_C^2 at broadside is the furthest from the radar system and has the smallest RCS of -13.5 dBsm. CR_B^3 and CR_B^1 both with a RCS of -1.5 dBsm are located at a relative angle to the radar system of 30° respectively -30° . The aim is to show the advantage of differently distributing the TX power to multiple targets of interest in order to increase the overall SNR. This measuring arrangement simulates a similar situation as shown in Fig. 1 with a far off located car (CR_C^2) in front of the radar system and a pedestrian (CR_B^3) at 30° . The obstacle (CR_B^1) at -30° was used to show the systems capability to suppress unwanted target responses. In FDMB MIMO operation, two beams are generated, a high-powered one ($v[1]=0.9$, $f_M[2]=1$ MHz), which is steered to broadside in order to illuminate CR_C^2 and a low-powered one ($v[2]=0.1$, $f_M[2]=2$ MHz) steered to 30° to illuminate CR_B^3 .

Spectra (Hann window applied) of the target responses, calculated from digitally down-converted IF signals of one RX channel, in FDMB MIMO as well as in FDMA MIMO operation modes are shown in Fig. 14. The x-axis in Fig. 14 is scaled to range. The peak magnitudes of the target responses are used to estimate the differences in received power, indicated by $\Delta P_{2.9m}=8$ dB, $\Delta P_{4.0m}=11.5$ dB, and $\Delta P_{2.2m}=8$ dB and associated to targets CR_B^1 , CR_C^2 ,

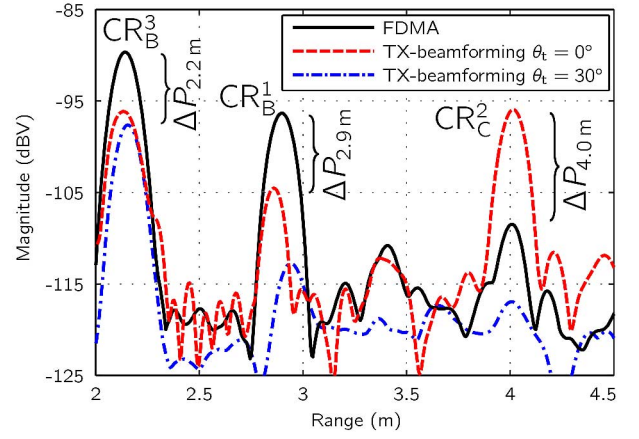


Fig. 14. Comparison of the received power levels of one RX channel between FDMB MIMO and FDMA MIMO operation.

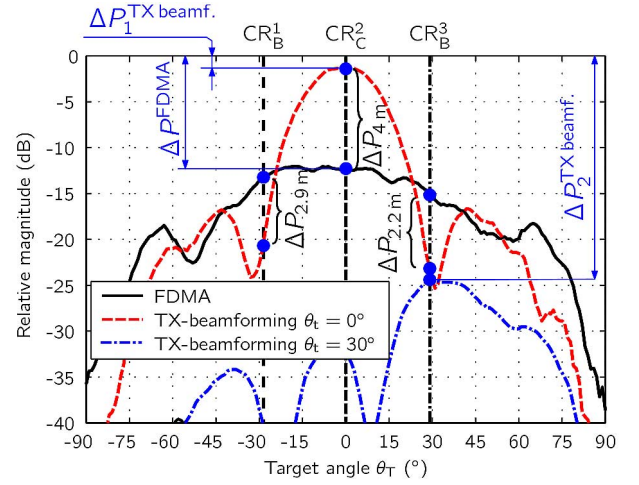


Fig. 15. Evaluation of measured differences in received power ($\Delta P_{2.2m}=8$ dB, $\Delta P_{2.9m}=8$ dB, and $\Delta P_{4.0m}=11.5$ dB) at corresponding angular position on the basis of two-way beam pattern recorded in Section V-B.

and CR_B^3 . These values correspond well with the measured two-way beam pattern as shown in Fig. 15. The beam pattern are scaled according to $\Delta P_{FDMA}=20 \log(1/M_t)=-12$ dB, $\Delta P_1^{TX \text{ beamf.}}=20 \log(v[1])=-0.9$ dB, and $\Delta P_2^{TX \text{ beamf.}}=20 \log(v[2])=-20$ dB. As can be seen the differences in received power agree well with the expected values.

Applying delay and sum beamforming to the measured IF signals in FDMB MIMO operation for $\theta_t[1]=\theta_{DBF}[1]=0^\circ$ and analogous in FDMA MIMO operation for $\theta_{DBF}=0^\circ$ results in spectra (Hann window function in range and rectangular in cross-range for the furthest target) as shown in Fig. 16. Two observations can be made, which also agree with theory. First, the DBF magnitude difference ΔP_{DBF} between the different operation modes is equal to $\Delta P_1^{TX \text{ beamf.}}$ as defined and shown in Fig. 14 and can be calculated with $\Delta P_{DBF} = \Delta P_{FDMA} + 20 \log M_t + \Delta P_1^{TX \text{ beamf.}}$. Second, the noise level drops in FDMB MIMO operation. As the dominating source of noise originates of the IF signal chain components, e.g., the IF filtering, amplification, and analog-to-digital conversion, the noise of all virtual FDMA channels sums up. The drop

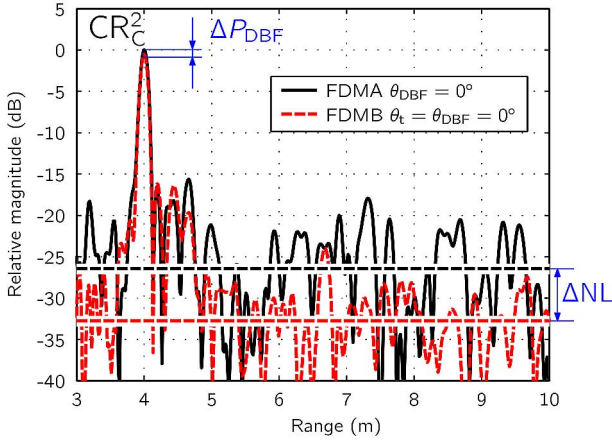


Fig. 16. DBF result at $\theta_{\text{DBF}}=0^\circ$ with theoretical drop of noise level $\Delta\text{NL}=6$ dB.

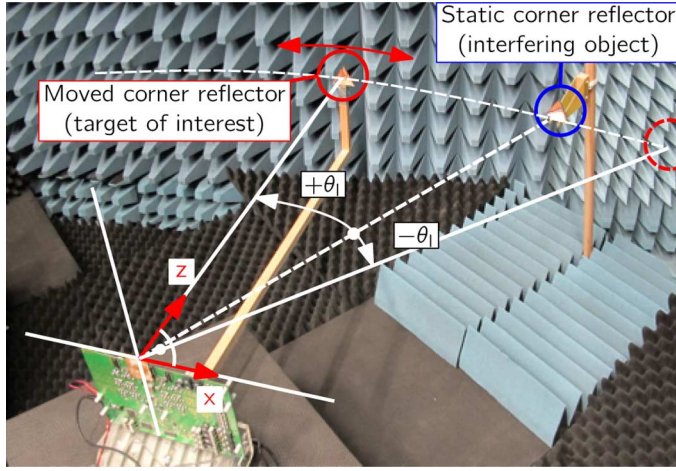


Fig. 17. Measuring arrangement used for angular measurements concerning the angular resolution and susceptibility in case of an interfering target.

in noise level ΔNL with $M_r^{\text{FDMA}}=M_r M_t$ and $M_r^{\text{FDMB}}=M_r$ can be stated as

$$\Delta\text{NL} = 10 \log(M_r^{\text{FDMA}}) - 10 \log(M_r^{\text{FDMB}}). \quad (16)$$

Here, it is assumed that the noise power density is equal for each IF channel, operation mode, and over the whole frequency range. The estimated difference in noise level as shown in Fig. 16 agree exact with the calculated value $\Delta\text{NL}=6$ dB. As a result, the increase of overall SNR in FDMB operation is obtained by the combination of the lower number of RX channels used for DBF and thus a lower resulting noise level and the increased TX power achieved by analog beamforming.

The second measurement arrangement is shown in Fig. 17. The aim is to conduct FDMB as well as FDMA MIMO measurements, to point out their susceptibility of angle estimation in case of an interfering target located at the same range and thus the same frequency. It will be shown that the advantage of the higher angular resolution in FDMA MIMO operation mode using a rectangular window function in cross-range leads to higher susceptibility of angle estimation. To get the same performance as in FDMB MIMO operation, window functions

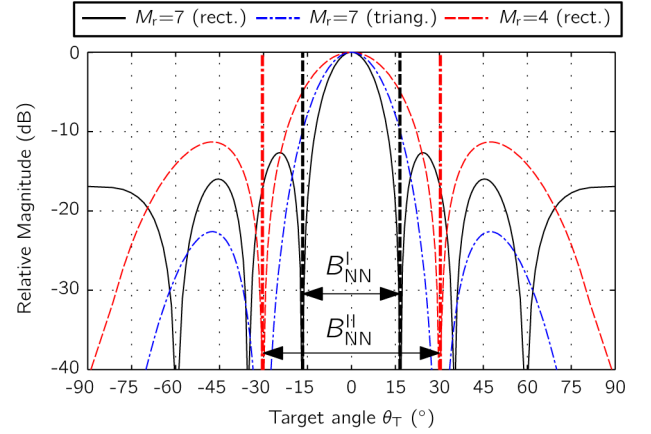


Fig. 18. RX beam patterns in FDMB MIMO ($M_r=4$, rectangular window function) and in FDMA MIMO ($M_r=7$, rectangular and triangular window function) operation mode applied in cross-range.

with lower sidelobes have to be used, with the consequence of a reduced angular resolution. Therefore, a corner reflector was additionally mounted on the rotary stage at a fixed distance to the RF frontend at broadside. The radar system as well as one corner reflector are rotated with respect to a static corner reflector at equal distance to the RF frontend. The latter is used as interfering target at angle θ_I . The angular position of θ_T (at broadside) as well as θ_I (varied from -90° to 90°) are referenced to the rotated coordinate system of the radar system. Measurements are conducted in FDMB MIMO operation as well as FDMA MIMO operation mode with $K=1$, $f_M[1]=1$ MHz, $v[1]=1$, and $\theta_t[1]=0^\circ$. The angular position of the corner reflector mounted to the rotary stage is estimated using methods of DBF and is denoted as $\hat{\theta}_T$. In Fig. 18 three RX beam patterns are shown. In case of FDMA MIMO operation mode $M_r=7$ considers the additional virtual antennas. Therefore, applying a rectangular window function in cross-range results in a higher achievable angular resolution in the FDMA MIMO than in the FDMB MIMO operation mode. The angular resolution, which is defined as the minimum distance for resolving two targets, is equal to the half of the null-to-null beamwidth achieved with a certain windowing function [28]. In Fig. 18 B_{NN}^{I} denotes the achievable null-to-null beamwidth using a rectangular window function in FDMA MIMO operation and $B_{\text{NN}}^{\text{II}}$ in case of using a triangular window function in FDMA and a rectangular window function in the FDMB MIMO operation mode. In FDMA MIMO operation mode, the amount of deviation between $\hat{\theta}_T$ and θ_T caused by an interfering target depends only on the used RX window function.

In FDMB MIMO operation mode, in which beamforming is also applied to the TX signals, $\hat{\theta}_T$ depends also on the TX beam pattern. The combination of a rectangular RX window function and uniform TX beam pattern results in a susceptibility to interference of angle estimation, which is comparable, but not equal, to that achieved with a triangular window function used in FDMA MIMO operation mode. The differences can be explained by examination of (2) and (9). In the presented system a discrete-time Fourier transform is used in cross-range to estimate the angular position $\hat{\theta}_T$. This is equivalent to

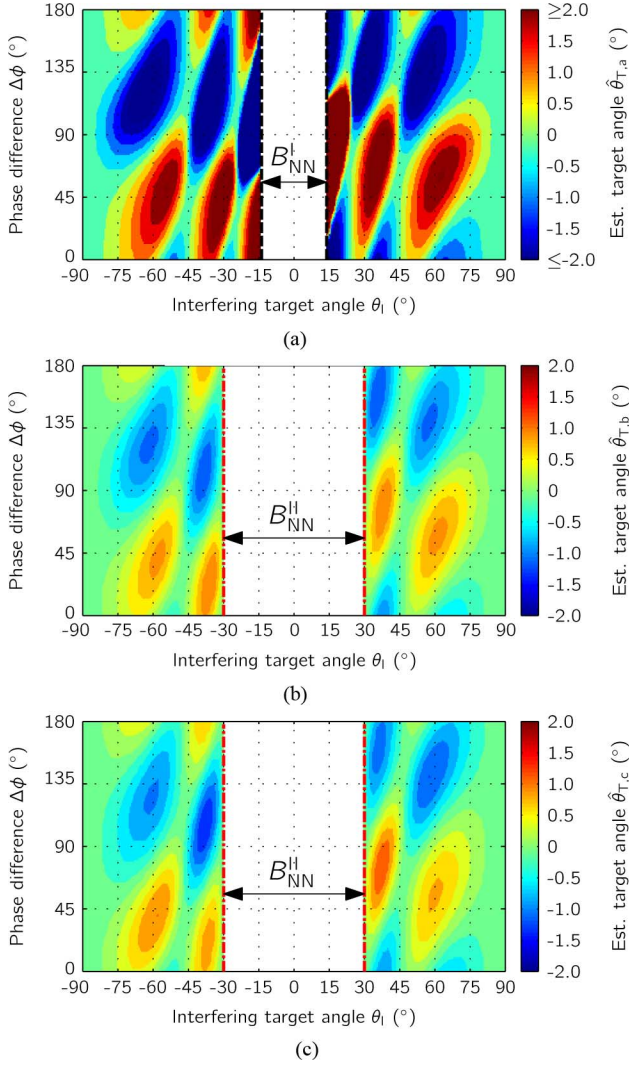


Fig. 19. Simulation of estimated angular position $\hat{\theta}_T$ of target at broadside interfered by a target at θ_I for phase differences between the two targets of $\Delta\phi$. Used window functions for DBF: (a) FDMA: $M_r=7$, rectangular. (b) FDMA: $M_r=7$, triangular. (c) FDMB: $M_r=4$, rectangular.

steering θ_{DBF} in (9) from -90° to 90° within one radar measurement. The angular position of the maximum peak magnitude is used to estimate θ_T . The steering angles $\theta_t[k]$ in (2) remain unchanged during one radar measurement. Thus, the overall beam pattern cannot be calculated by the convolution of digital RX and analog TX beam patterns. Combining multiple measurements with different $\theta_t[k]$ directed into the direction of the target would lead to an overall beam pattern equal to the convolution. However, this kind of operation is also possible with the presented system but is not scope of this work.

To clarify this matter, simulation results for $\hat{\theta}_T$ of two interfering targets are shown in Fig. 19. $\theta_T=0^\circ$ and θ_I is varied from -90° to 90° . In order to account for unavoidable inaccuracies in the measurement arrangement, simulations are performed for phase differences $\Delta\phi$ in the range of 0° to 180° between the received target responses. By comparing Figs. 19(a) and 19(b) it can be seen that the advantage of the narrow beam achieved with a rectangular window function comes at the cost of increased susceptibility of angle estimation in case of an inter-

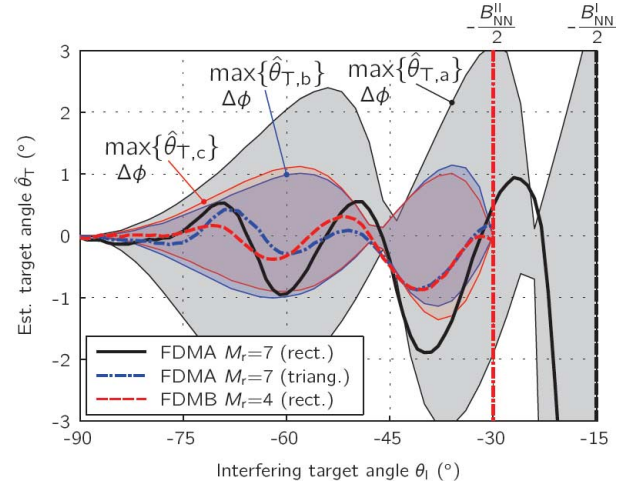


Fig. 20. A target (RCS= -1.5 dBsm) at broadside is interfered by a target (RCS= -1.5 dBsm) at θ_I for values of -90° to -15° . Depicted are the estimated angular position $\hat{\theta}_T$ determined from measurements and worst case boundaries, which are the envelopes of the projection of plots in Figs. 19(a)–19(c) on the $\theta_I/\hat{\theta}_T$ -plane.

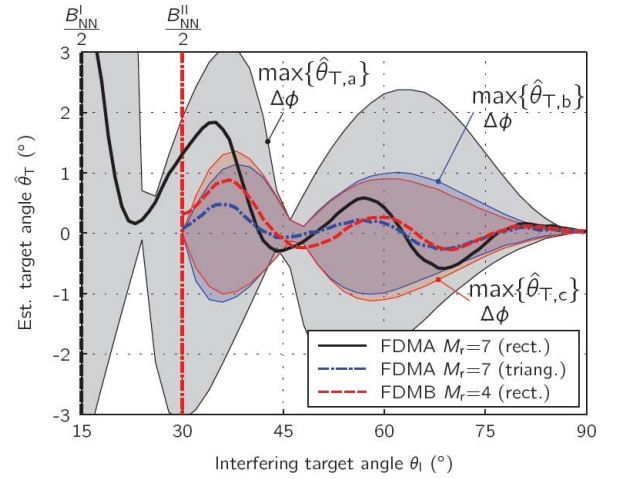


Fig. 21. A target (RCS= -1.5 dBsm) at broadside is interfered by a target (RCS= -1.5 dBsm) at θ_I for values of 15° to 90° . Depicted are the estimated angular position $\hat{\theta}_T$ determined from measurements and worst case boundaries, which are the envelopes of the projection of plots in Figs. 19(a)–19(c) on the $\theta_I/\hat{\theta}_T$ -plane.

fering target. Even for θ_I outside $\pm 1/2 B_{NN}^I$, the deviations are two times higher than using a triangular window function with a lower sidelobe level (SLL). This effect is caused by the spectral leakage, which varies for different window functions [29]. As already mentioned, the simulated values of deviation (are equal to $\hat{\theta}_T$ for $\theta_T=0^\circ$) for FDMA MIMO employing a triangular window function and FDMB MIMO employing a rectangular window function for RX and TX are in the same range as it can be seen by comparing Fig. 19(b) with Fig. 19(c). Envelopes of the projection of plots shown in Figs. 19(a)–19(c) on the $\theta_I/\hat{\theta}_T$ -plane are plotted in Figs. 20 and 21 in order to allow a evaluation of the measurement results. Even though inaccuracies in the measurement arrangement, the DBF results of $\hat{\theta}_T$ obtained with measurements are in good accordance with theory.

Worst case scenarios of interfering targets located within the same DBF region do not differ significantly from those in radar

systems without modulation. To avoid cases in which targets lie outside the DBF region and appear as ghost targets in the next spectral DBF region, care must be taken in offset frequency planning. Adaptively extending the spectral DBF region for high power TX beams can be used to avoid such sources of interferences.

VI. CONCLUSION

In this work we presented a prototype 77-GHz FMCW MIMO radar system that employs orthogonal frequency waveforms to generate simultaneously multiple TX beams for tracking multiple targets in automotive applications. The presented principle allows standard beam pattern synthesis for each beam, which simplifies the design procedure. Measured beam pattern of multiple simultaneously generated beams are in good accordance with simulation, based on theoretical beam pattern synthesis in combination with measured single antenna element beam pattern.

Key and limiting components in the presented system are the VMs in each TX path of the PA antenna system fed by DACs. In order to achieve a higher TX power CF minimization concerning the VM's control signals was discussed. In this way the transmitted power could be increased by 5 dB as shown in case of a scenario with five simultaneously generated beams.

It has been also shown that weighting the TX power between different beams can improve the performance in terms of SNR compared to MIMO systems without TX beamforming as shown in measurements. Furthermore the dependency of angular resolution on the beam pattern was discussed, based on measurements conducted in FDDB MIMO as well as in FDMA MIMO operation mode using the same RF frontend. Additionally, the susceptibility of angle estimation in case of an interfering target located at the same range was investigated.

The presented system achieved the expected performance, as predicted by theory and verified by measurements. Although, the radar system employs only four TRX antennas the proposed principle and the potential benefits were shown. The presented approach offers additional degrees of freedom concerning overall beam pattern, especially for applications with a limited number of antennas. Expanding the array size is possible, however, the following points must be considered and observed. How many beams are required and does the baseband hardware support the required bandwidth for the desired unambiguous ranges? What are the requirements on the beam shape? Is the increase of bandwidth due to modulation still compliant with regulations? Admittedly, the presented approach won't replace classical radar systems, but represents a step towards software defined or cognitive radar systems. Therefore, the proposed FDDB MIMO radar system may serve as a basis for future designs of automotive radar sensors.

REFERENCES

- [1] A. G. Stove, "Linear FMCW radar techniques," *Proc. Inst. Elect. Eng. — Radar Signal Process.*, vol. 139, no. 5, pt. F, pp. 343–350, Oct. 1992.
- [2] J. Hasch, E. Topak, R. Schnabel, T. Zwick, R. Weigel, and C. Waldschmidt, "Millimeter-wave technology for automotive radar sensors in the 77 GHz frequency band," *IEEE Trans. Microw. Theory Tech.*, vol. 60, no. 3, pp. 845–860, Mar. 2012.
- [3] H. Li and B. Himed, "Transmit subaperturing for MIMO radars with co-located antennas," *IEEE J. Sel. Topics Signal Process.*, vol. 4, no. 1, pp. 55–65, 2010.
- [4] E. Brookner, "Mimo radar: Demystified," *Microw. J.*, vol. 56, no. 1, pp. 22–44, Jan. 2013.
- [5] D. R. Culkun and T. D. Graham, "US Patent 8,289,203 B2," 2009.
- [6] C. Pfeffer, R. Feger, C. M. Schmid, C. Wagner, and A. Stelzer, "An IQ-modulator based heterodyne 77-GHz FMCW colocated MIMO radar system," in *Proc. IEEE MTT-S Int. Microwave Symp. Digest (MTT)*, Jun. 2012, pp. 1–3.
- [7] J. J. M. de Wit, W. L. van Rossum, and A. J. de Jong, "Orthogonal waveforms for FMCW MIMO radar," in *Proc. IEEE Radar Conf.*, 2011, pp. 686–691.
- [8] B. Friedlander, "On transmit beamforming for MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 4, pp. 3376–3388, Oct. 2012.
- [9] C. Pfeffer, R. Feger, C. Wagner, and A. Stelzer, "A multimode-beamforming 77-GHz FMCW radar system," in *Proc. IEEE MTT-S Int. Microwave Symp. Digest (MTT)*, Jun. 2013, pp. 1–4.
- [10] M. Steinhauer, H.-O. Ruo, H. Irion, and W. Menzel, "Millimeter-wave-radar sensor based on a transceiver array for automotive applications," *IEEE Trans. Microw. Theory Tech.*, vol. 56, no. 2, pp. 261–269, 2008.
- [11] P. Sammartino, C. Baker, and H. Griffiths, "Frequency diverse MIMO techniques for radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 49, no. 1, pp. 201–222, 2013.
- [12] D. W. Bliss and K. W. Forsythe, "Multiple-input multiple-output (MIMO) radar and imaging: degrees of freedom and resolution," in *Proc. Rec. 37th Asilomar Conf. Signals, Syst. Comput.*, 2003, pp. 54–59.
- [13] J. Li and P. Stoica, "MIMO radar with colocated antennas," *IEEE Signal Process. Mag.*, vol. 24, no. 5, pp. 106–114, Sep. 2007.
- [14] C. Wagner, A. Stelzer, and H. Jäger, "A phased-array radar transmitter based on 77-GHz cascaded transceivers," in *Proc. IEEE MTT-S Int. Microw. Symp. Digest*, Jun. 2009, pp. 73–76.
- [15] R. Feger, C. Wagner, and A. Stelzer, "An IQ-modulator based heterodyne 77-GHz FMCW radar," in *Proc. German Microw. Conf. (GeMIC)*, Mar. 2011, pp. 1–4.
- [16] R. Feger, C. Wagner, S. Schuster, S. Scheibhofer, H. Jäger, and A. Stelzer, "A 77-GHz FMCW MIMO radar based on an SiGe single-chip transceiver," *IEEE Trans. Microw. Theory Tech.*, vol. 57, no. 5, pp. 1020–1035, May 2009.
- [17] S. Schuster, S. Scheibhofer, L. Reindl, and A. Stelzer, "Performance evaluation of algorithms for SAW-based temperature measurement," *IEEE Trans. Ultrason., Ferroelectr., Freq. Contr.*, vol. 53, no. 6, pp. 1177–1185, Jun. 2006.
- [18] R. C. Hansen, *Phased Array Antennas*, K. Chang, Ed., 2nd ed. New York, NY, USA: Wiley, 2009.
- [19] M. Friese, "Multitone signals with low crest factor," *IEEE Trans. Commun.*, vol. 45, no. 10, pp. 1338–1344, Oct. 1997.
- [20] M. Solomou, C. Evans, and D. Rees, "Crest factor minimisation in the frequency domain," in *Proc. 18th IEEE Instrum. Meas. Technol. Conf.*, 2001, vol. 2, pp. 1375–1381.
- [21] M. Schroeder, "Synthesis of low-peak-factor signals and binary sequences with low autocorrelation (corresp.)," *IEEE Trans. Inf. Theory*, vol. 16, no. 1, pp. 85–89, Jan. 1970.
- [22] C. Wagner, M. Hartmann, A. Stelzer, and H. Jaeger, "A fully differential 77-GHz active iq modulator in a silicon-germanium technology," *IEEE Microw. Wireless Compon. Lett.*, vol. 18, no. 5, pp. 362–364, May 2008.
- [23] R. Feger, C. Wagner, and A. Stelzer, "A W-band heterodyne FMCW radar based on TX IQ-modulation," *Frequenz*, vol. 65, no. 7–8, pp. 183–191, Aug. 2011.
- [24] Z. Tong, C. Wagner, R. Feger, A. Stelzer, and E. Kolmhofer, "A novel differential microstrip patch antenna and array at 79 GHz," in *Proc. Int. Symp. Antennas Propag.*, Taipei, CHN, Taiwan, 2008.
- [25] C. Wagner, A. Stelzer, and H. Jäger, "PLL architecture for 77-GHz FMCW radar systems with highly-linear ultra-wideband frequency sweeps," in *Proc. 2006 IEEE MTT-S Int. Microwave Symp. Dig.*, San Francisco, CA, USA, Jun. 2006, pp. 399–402.
- [26] C. Pfeffer, R. Feger, C. M. Schmid, A. Fischer, and A. Stelzer, "A multi-channel baseband board for mm-Wave radar applications," in *Proc. Microelectron. Conf.*, Vienna, AT, Apr. 2012, pp. 65–70.
- [27] S. Y. Kim, O. Inac, C.-Y. Kim, and G. Rebeiz, "A 76–84 GHz 16-element phased array receiver with a chip-level built-in-self-test system," in *Proc. Radio Freq. Integr. Circuits Symp. (RFIC)*, 2012 IEEE, Jun. 2012, pp. 127–130.
- [28] H. L. v. Trees, *Detection, Estimation, and Modulation Theory, Part IV, Optimum Array Processing*. New York, NY, USA: Wiley Interscience, 2002.
- [29] A. V. Oppenheim, R. W. Schaffer, and J. R. Buck, *Discrete-Time Signal Processing*, 2nd ed. Englewood Cliffs, NJ, USA: Prentice Hall, 1999.



Clemens Pfeffer (S'11) was born in St.Pölten, Austria, in 1975. He received the Dipl.-Ing. (M.Sc.) degree in mechatronics from Johannes Kepler University, Linz, Austria, in 2009.

In 2009, he joined the Institute for Communications and Information Engineering, Johannes Kepler University, as a Research Assistant and became a member of the Christian Doppler Laboratory for Integrated Radar Sensors, Johannes Kepler University. His research interests are radar system design for industrial and automotive radar sensors.

Mr. Pfeffer was recipient of the 2012 European Conference on Antennas and Propagation (EuCAP) Best Measurement Paper Prize.



Reinhard Feger (S'08–M'11) was born in Kufstein, Austria, in 1980. He received the Dipl.-Ing. (M.Sc.) degree in mechatronics and Dr. techn. (Ph.D.) degree in mechatronics from Johannes Kepler University, Linz, Austria, in 2005 and 2010, respectively.

In 2005, he joined the Institute for Communications and Information Engineering, Johannes Kepler University, as a Research Assistant. In 2007, he became a member of the Christian Doppler Laboratory for Integrated Radar Sensors, Johannes Kepler University. His research topics are radar

signal processing, as well as radar system design for industrial and automotive radar sensors.

Dr. Feger was recipient of the 2011 Microwave Prize and the 2011 German Microwave Conference Best Paper Award. In 2012 he received the Best Measurement Paper Prize at the European Conference on Antennas and Propagation.



Christoph Wagner (S'06–M'11) was born in Zwettl, Austria, in 1980. He received the Dipl.-Ing. (M.Sc.) degree in mechatronics and Dr. techn. (Ph.D.) degree in mechatronics from the Johannes Kepler University, Linz, Austria, in 2006 and 2010, respectively.

In 2006, he joined Danube Integrated Circuit Engineering (DICE) in Linz, Austria. In 2007, he became a member of the Christian Doppler Laboratory for Integrated Radar Sensors. Since 2010, he is with DICE again, where he is currently responsible for the development of 77-GHz integrated radar transceivers.

His research interests include circuit and system design for millimeter-wave and microwave radar sensors, with emphasis on digitally intensive techniques for RF imperfection mitigation. He has authored or co-authored more than 30 journal and conference papers. He serves a reviewer for international journals and conferences.

Dr. Wagner is the recipient of the 2011 IEEE Microwave Prize, the 2011 award of the Information Technology society of the Electrotechnical Association (OVE-GIT), as well as a co-recipient of the best paper award at the German Microwave Conference 2011 and the 62nd ECTC outstanding paper award. Christoph Wagner is a member of the IEEE and member of the Microwave Theory and Techniques, as well as the Solid-State Circuits society.



Andreas Stelzer (M'00) was born in Haslach an der Mühl, Austria, in 1968. He received the Diploma Engineer degree in electrical engineering from the Technical University of Vienna, Vienna, Austria, in 1994, and the Dr. techn. degree (Ph.D.) in mechatronics (with *sub auspiciis praesidentis rei publicae* honors) from the Johannes Kepler University, Linz, Austria, in 2000.

In 2003, he became an Associate Professor with the Institute for Communications and Information Engineering and RF Systems, Johannes Kepler University. Since 2003, he has been a key Researcher for the Linz Center of Competence in Mechatronics (LCM) and currently for the 2008 founded Austrian Center of Competence in Mechatronics (ACCM), where he is responsible for numerous industrial projects. Since 2007, he has been Head of the Christian Doppler Laboratory for Integrated Radar Sensors, and since 2011 he is full Professor at Johannes Kepler University, heading the department for RF-Systems. He has authored or coauthored over 245 journal and conference papers. He is a Reviewer for international journals and conferences. His research is focused on microwave sensor systems for industrial and automotive applications, RF and microwave subsystems, surface acoustic wave (SAW) sensor systems and applications, as well as digital signal processing for sensor signal evaluation.

Dr. Stelzer is a member of the Austrian ÖVE. He has served as an associate editor for the IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS. He was the recipient of several awards including the IEEE COM Innovation Award and the European Microwave Association (EuMA) Radar Prize of the European Radar Conference. He was also the recipient of the 2011 German Microwave Conference (GeMiC) Best Paper Award as well as of the 2008 IEEE Microwave Theory and Techniques Society (IEEE MTT-S) Outstanding Young Engineer Award and the 2011 IEEE Microwave Prize. In 2012 he received the Best Measurement Paper Prize Award at the European Conference on Antennas and Propagation and the Best Paper Award at the Asia-Pacific Conference on Antennas and Propagation.